

System of Linear Equations

Time limit: 1.00 s

Memory limit: 512 MB

You are given $n \cdot (m + 1)$ coefficients $a_{i,j}$ and b_i which form the following n linear equations:

- $a_{1,1}x_1 + a_{1,2}x_2 + \dots + a_{1,m}x_m = b_1 \pmod{10^9 + 7}$
- $a_{2,1}x_1 + a_{2,2}x_2 + \dots + a_{2,m}x_m = b_2 \pmod{10^9 + 7}$
- ...
- $a_{n,1}x_1 + a_{n,2}x_2 + \dots + a_{n,m}x_m = b_n \pmod{10^9 + 7}$

Your task is to find any m integers $x_1, x_2, \dots, x_m$ that satisfy the given equations.

Input

The first line has two integers n and m : the number of equations and variables.

The next n lines each have $m + 1$ integers $a_{i,1}, a_{i,2}, \dots, a_{i,m}, b_i$: the coefficients of the i -th equation.

Output

Print m integers $x_1, x_2, \dots, x_m$: the values of the variables that satisfy the equations. The values must also satisfy $0 \leq x_i < 10^9 + 7$. You can print any valid solution. If no solution exists print only -1 .

Constraints

- $1 \leq n, m \leq 500$
- $0 \leq a_{i,j}, b_i < 10^9 + 7$

Example

Input	Output
3 3 2 0 1 7 1 2 0 0 1 3 1 2	2 1000000006 3